

Examiner reports

Q1.

In part (a), most correctly read the graph and then correctly used the coordinates to give the required P and V values.

The majority seemed well practised in the first step for part (b), writing the equation as $\log_{10}P = \log_{10}C + d\log_{10}V$. Many progressed to a correct solution easily and completed the question. Others became confused about what was a log and what was not or failed to handle the negative values and negative gradient. Those who tried to use point A and another point to generate simultaneous equations usually became lost at some point in the algebra.

In part (c), most remembered to include the units, relating the mathematical model back to the physical situation behind it.

Q2.

This question required the use of the exponential function and natural logs which produced a diverse range of responses. In part (a) about a third of students were able to correctly write down the required two equations using the information given in the question, but solving them simultaneously proved beyond many students. Some used the given value of b to find the value of a , which gained 1 mark and it also enabled them to tackle later parts of the question.

There were many correct responses to part (b) which was well done as long as students had found the correct values of a and b .

In part (c) students mostly failed to use inequalities but this approach did not lose marks. Those students who isolated the exponential term before taking logs were usually more successful than those who took logs straight away, as errors occurred later when simplifying the log terms. Many lost the final mark because they did not quote a year as their final answer as required. Appropriate numerical approaches were accepted and there was some follow through of the students value of ' a ', provided $b = 90$ was used.

There were some very good responses to part (d) and over half of all students were able to give a limitation of the model. It was disappointing that over 17% of students did not attempt this question despite it not being dependent upon any of the earlier work in the question.

Q3.

This question was answered well by most students. Many scored 2 out of a possible 3 marks and could find the inverse function and state it in a correct form. The most frequent mistake seen was incorrectly stating the domain or missing it out altogether. The mark for the correct domain was independent and could be achieved without correctly finding $f^{-1}(x)$, as it could be deduced from the range of $f(x)$.

Q4.

Part (a) split students into two groups. Almost equal numbers scored full marks or no

marks. Finding the gradient of a curve given in parametric form should be a well-practiced and routine technique. However, differentiating 2^t is new to this specification and this caught out many students.

Part (b) was completed more successfully. Students who took an approach similar to the typical solution given in the mark scheme, tended to complete the question and often scored full marks. Students who introduced logs, to make t the subject, scored the first mark for choosing a correct strategy. These students were often unable to manipulate their logs and failed to complete the argument correctly, which is what the third mark was for.

Q11.

In part (a) a large proportion of the students recognised the transformation as a stretch although a significant minority incorrectly stated that it was in the x -direction with scale factor $1/3$. In part (b) there were many correct sketches although it was not uncommon to see the

y - intercept incorrectly given as 1. In part (c), the majority of students scored the first mark by eliminating y correctly to obtain $4^{-x} = 3 \times 4^x$ although a significant minority failed to score even this opening mark because they tried to combine two steps which resulted in the incorrect initial equation, $-x \log 4 = x \log 12$. A significant number of the students failed to make any further progress. More able students who used the log laws correctly to reach $-x \log 4 = \log 3 + x \log 4$ were generally successful in finding the correct value for x . A common error in applying laws of indices, amongst those students who tried to rearrange the equation $4^{-x} = 3 \times 4^x$ before taking logarithms, is illustrated by the incorrect equation

$$16^{2x} = \frac{1}{3}.$$

Q12.

The range of marks seen in this question typically varied between up to 4, scored by about 85% of the students, to up to 7 scored by about 25%. Very few completed the whole question successfully, with many making only superficial attempts at part (b).

Part (a)(i) and (ii). Most students answered correctly with only the occasional arithmetic error.

Part (a)(iii). Many students scored 2 out of the 3 marks available; although they understood what to do and followed correct manipulation, taking logarithms correctly, they didn't respond to the request for integers in the final expression. Those students who didn't simplify the expressions in their working often made an error.

Part (b)(i). Students were expected to attempt the required differentiation using the quotient or chain rule. Those that did this often made a sign error or had a wrong coefficient, sometimes including a t in their expression. Although some excellent algebra was seen in deriving the given result using the initial expression for N , many were unsuccessful in eliminating the $9e^{\frac{1}{3}}$ term, and some gave up without trying. However, many students tried to rearrange the given equation for N to make $e^{\frac{1}{3}}$ or t the subject, and then differentiate. It is difficult to see why they considered this approach would be helpful, but many made an algebraic error and little progress. Some tried to solve the differential equation and made little or no progress.

Part (b)(ii). Most students who attempted this understood the question to mean finding the maximum value of N rather than that of the growth rate $\frac{dN}{dt}$, but most had to abandon their attempt. Of those who did differentiate the growth rate expression, most wrote $\frac{d^2N}{dt^2}$, which was condoned; if their attempt led to the maximum growth rate occurring when $N = 250$ they usually completed successfully; those who had worked with inequalities were condoned. Some students thought $T = 250$ was the required answer. Few students saw that the maximum value of the growth rate was at $N = 250$ by inspection, (due to the symmetries of the quadratic function involved) but some concise and correct solutions to the problem were seen.

Q13.

The sketch of the given exponential curve was usually drawn having the correct intercept at $y = 1$, but a significant minority of students did not show its correct behaviour for negative x values – the curve should level out close to the x -axis and certainly nearer to the x -axis than the line $y = 1$. Part (b) was usually answered correctly with almost all showing that they knew how to use logarithms to solve an equation of the form $a^x = b$. The positioning of the minus sign was crucial in part (c) and it was not uncommon to see the incorrect answer $f(x) = -9^x$.

Q14.

This question, which tested the 'exponentials and logarithms' section of the specification, resulted in a good spread of marks. Although there were many correct sketches of $y = \frac{1}{2^x}$ seen in solutions to part (a), there were almost as many that were sketches similar to that of $y = 2^x$. Another common incorrect sketch was one which was obtained by reflecting $y = 2^x$ in the x -axis.

Part (b) was generally answered well, although some candidates failed to pick up the final mark because they either left their final answer in the form $-x = 0.322$ or did not give the value of x to at least the required degree of accuracy. Of those who were less successful, some displayed poor algebraic skills or believed incorrectly that cross multiplying led to an equation of the form $10^x = 4$.

Part (c) proved to be demanding, with most candidates failing to obtain the correct expression for y in terms of a and b , although the majority of candidates were awarded partial credit for applying log laws correctly. A typical misconception is illustrated by $2 \log_a b + 3 \log_a y = 5 \log_a by$. Those candidates who were the most successful usually applied no more than one log law (or used $\log_a a = 1$) per line of working to reach, for example, $\log_a b^2 y = 1$ and then stated $b^2 y = a$ before completing by rearranging to obtain $y = ab^{-2}$.

Q15.

Very few candidates scored full marks for this question.

Part (a) was well done with only a handful failing to recognise a stretch and most giving the correct direction and scale factor; most who were wrong in one of these were wrong in both. It was essential to use the correct term 'translate' for the other, so 'shift' did not earn marks and 'transformation' was inadequate. A few made an error in the vector but most were correct.

Most candidates were unsuccessful in answering part (b) of the question as their curve was either below the x -axis or continued into the second quadrant. Many placed the curve through e instead of $e + 1$, and the curvature, for $x < e + 1$, was often wrong.

Correct answers in part (c)(i) were sometimes spoiled by poor algebra. It was worrying to see the misunderstanding of the log function, with $\ln(x - e)$ expanded as $\ln x - \ln e$ and incorrect subsequent work once a correct answer had been found: $e^{-1} + e = 0$ was a common such response.

In part (c)(ii), although the first mark for $x \geq 2e$ was often earned (though $x \leq 2e$ was common), the second mark was only gained by a very few candidates, as the part of the inequality concerning e was generally omitted completely.

Q16.

Most candidates demonstrated some understanding of the information given in this in-context question on exponential growth and decay. There were many fully correct solutions.

Most candidates had part (a) correct, with relatively few either not converting 1 hour to 60 minutes or failing to round correctly to the nearest 1000, or not rounding at all.

In part (b)(i), most candidates set up the correct starting-point equation, but they were required to manipulate it to a point where logarithms could be taken before marks were awarded; many failed to do this. Knowledge of using logarithms varied widely. Many were unable to deal properly with the coefficient and the exponential term and wrote down equations in which t actually should have cancelled out. However, many candidates managed to reach a 'solution' from such incorrect working. In contrast, many other candidates gave a concise derivation of the correct answer.

In part (b)(ii), again many candidates set up the correct starting equation, or something close to it, and some used inequalities, which were accepted. Although many did show clearly how the given equation was derived, others could not handle the exponential terms correctly or made sign errors in manipulating their equation. However, many of those who could not derive the quadratic equation in $e^{\frac{1}{3}t}$ did recognise it as such and solved it successfully.

Pleasingly, most candidates also rejected the $e^{\frac{1}{3}t} = -10$ as impossible. Relatively few tried to find a solution again via taking logarithms immediately, but those who did made no progress and abandoned the question.

Q17.

The sketch of the given exponential curve was generally well-done although some students did not display sufficient care in sketching the curve in the negative- x region. In part (b), the majority of students were awarded the mark for correctly eliminating one of

the variables, but a significant number could not form a quadratic with any confidence. Those who recognised that 7^{2x} could be written as $(7^x)^2$ usually had no difficulty in proceeding to form and solve a correct quadratic equation in y and then to find the correct value for the y -coordinate of the point of intersection, although a small minority of these students did not give sufficient consideration to the rejection of the inappropriate negative solution. The usual errors seen are illustrated by the following: $7^x = 7^{2x} - 12 \Rightarrow 12 = 7^{2x} - 7^x \Rightarrow 12 = 7^x$ or $7^x = 7^{2x} - 12 \Rightarrow \log 7^x = \log 7^{2x} - \log 12$. There were, however, an encouraging number of completely correct solutions.

Q18.

Most students did part (a)(i) correctly, with very few failing to round to the requested nearest £10. Some confused P and V , so the value of the investment actually went down.

For part (a)(ii), most students showed that they understood the information given and wrote down a correct opening expression, although there was uncertainty over which way the inequality should go. Some then simplified and solved using logarithms efficiently to get the correct answer. Some resorted to trial and improvement, often getting to the correct answer, whilst some students tried a hybrid of both logs and trial and error. Some students failed to give their answer in complete years, thus losing the final mark.

In part (b) also, most students showed they understood the information given and wrote down a correct opening expression. Some made errors in simplifying, but many students who attempted a solution by logarithms, took logarithms incorrectly and their unknown (T

or n) cancelled out of their equation. A common error was to combine 1.5 from $\left(\frac{1500}{1000}\right)$ with $\left(1 + \frac{1.5}{100}\right)$ to get a term $n \log 1.5 \left(1 + \frac{1.5}{100}\right)$. Relatively few students got to the correct answer using logarithms. Some realised they had made an error but found the correct answer of 28 from somewhere; it was not always clear what they had done. However, many abandoned their attempt at this point. Many, though, did get a correct answer through trial and improvement, which was accepted for full marks if done correctly, but was worth no marks otherwise.

Q19.

Most of the candidates who used the expected algebraic approach successfully reached the required value for the common ratio. Very few candidates quoted incorrect formulae for the third and sixth terms. Those who assumed the value of r or just showed arithmetical working rarely gained any credit in part (a)(i). A large majority of candidates scored full marks in part (a)(ii) for finding the first term of the series. Most candidates realised that the sum of the first twenty terms of the series was required and so picked up the method mark, but errors such as ' $4 \times 3^{20} = 12^{20}$ ' in intermediate working towards the printed answer was enough to lose the accuracy mark. Part (b)(ii) was not answered well, with a significant majority failing to score even a single mark. A common error was to replace u_n by the formula for the sum of the first n terms. Those candidates who did make progress usually substituted a correct expression, $4 \times 3^{n-1}$, for u_n , and a high proportion of these candidates then took logarithms and used an appropriate logarithmic law correctly. However, a significant number of these candidates failed to score the final mark because they gave the non-integer value 32.43 for n or they rounded down to 32. Some candidates used a trial and improvement approach in preference to taking logarithms with varying degrees of success. When applying a trial and improvement approach, candidates should be aware that it is insufficient to just use one value of n .

Q20.

Most candidates scored well on this question although some misinterpreted the information given.

In part (a) most candidates substituted and evaluated correctly. A few, having written the unsimplified form, could not cope with the index and combined the 10 and 2 incorrectly. Some misinterpreted the information and found a mass that decayed to 10 grams in 14 days.

In part (b) most candidates substituted the 16 correctly and then, usually, took logs in order to

solve for d . Only a minority recognised that $\frac{1}{16} = 2^{-4}$ and so $\frac{d}{8} = 4$. Some candidates began

incorrectly by writing $2^{-\frac{d}{8}} = 16$ which led to a negative answer. Most of these then just dropped the minus sign rather than looking for an error in their working.

In (c) Most candidates set up the correct equation and used logarithms to various bases to solve it. The most common answer here was 53.15, as candidates failed to heed the request to give the answer as an integer. Many also rounded down to 53, misinterpreting what the question was asking.

Not all understood the question and assumed the mass had decayed by 1%, so used 99%.

Q21.

Graphs were drawn correctly by the majority of candidates and the correct vector given for the translation. There are still too many candidates, though, who use alternative words for 'translation' such as "move" or "shift" which are not accepted.

In part (c)(i) the justification of the transformation of the index equation was more of a challenge even to the high grade candidates. A substantial number of candidates simply replaced 4^x with $(2^x)^2$ which gained no credit as it was basically what they had to show. More were successful in showing that $2^{x+2} = 4^Y$ by inserting the intermediate stage $= 2^x \times 2^2$. The final part was only done well by better candidates with many misusing the laws of logs at the outset as illustrated by

' $\log 4^x - \log 2^{x+2} = \log 5$ '. Of those who did solve the printed quadratic equation in Y , equated the values to 2^x and solved for x , where possible, only a minority gave a conclusion for the "show that" that there was "only one real solution".

Q22.

Virtually all candidates had part (a)(i) correct.

In (a)(ii) over 80% of the candidates were correct here, although a variety of methods were seen, ranging from taking the 25th root of 25, to expressions involving logarithms. Those who failed to achieve both marks, usually started correctly, but did not continue to give an expression from which the correct value of k could be found.

In part (b) most candidates were able to use logarithms correctly to solve their equation

for t , with few using trial and improvement. However, many then interpreted the result incorrectly rounding down rather than up, not realising that as the result was over 55 it indicated the 56th year and thus 2016. Some candidates used the wrong base year, 1985 being common, and some didn't interpret their answer as a year at all.

Q23.

Candidates' sketches generally displayed a thorough understanding of the graph of $y = 2^x$. In part (b)(i) the trapezium rule was frequently applied correctly and a smaller proportion of candidates than normal failed to give their final answer to the stated degree of accuracy. Part (b)(ii) was almost always answered correctly. In part (c), the question asked for a description of a geometrical transformation. Although answers which gave two separate translations were condoned, the penalty for descriptions which involved two different transformations, for example stretch and translation, was severe. However many correct descriptions were presented, which indicates the continued improvement in candidates' performance on this topic. Part (d), as expected, proved to be demanding. The common wrong approach is illustrated by ' $\log y = \log 2^{x+k} + \log 3$ '. Better candidates applied the correct substitution and manipulation to reach $2^k = 5$, but a significant minority of these candidates, after taking logs and using the log law, left their answer in the form k

$$= \frac{\log 5}{\log 2}.$$

Q24.

Most candidates answered part (a)(i) correctly, although a surprising number seemed to think that $A \times 0 = A$.

Most candidates did part (a)(ii) convincingly, usually choosing to evaluate the exponential expression as 0.95 ... and solving for A to demonstrate its value is 60 to two significant figures. Those who did not actually solve for A or did not show A as 59.9...were penalised.

Most candidates started part (b)(i) well, showing they had interpreted the information given correctly. However many could not manipulate their resulting $\ln$ expression into the form required; many, either not knowing what was meant by " a and b are integers" or not taking notice of it, left their result in fractions. Many candidates 'lost' a minus sign during their working whilst some attempted to take the $\ln$ of a negative number.

In part (b)(ii), a few candidates were able to derive the result clearly and efficiently; others managed it after some working that was somewhat difficult to follow whilst others made an error such as dropping a minus sign but still claimed they had obtained the result.

However, many candidates gave up after finding $\frac{dA}{dt}$. A few candidates attempted to integrate the given differential equation, some successfully recovering the given expression for A , which was both acceptable and impressive.

Although many candidates had part (b)(iii) correct, many errors were also seen, quite a

common one being $4 \times 2 = \frac{1}{2}$. The other common error was to substitute $h = 13$, rather

than $\frac{dz}{dt} = 13$. It was notable that many candidates attempted part (b)(iii) before returning to try and sort out part (b)(ii), which was sensible.

Q25.

Parts (a) and (b) gave the opportunity for all to pick up marks, with the majority giving the two decimal place answer required in part (b). Many candidates scored just one of the two marks in part (c). The translation approach was the more popular, but the fact that the

vector needed to be $\begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$ was too subtle for most, as was the scale factor of $\frac{1}{8}$ for those using a stretch approach. Similarly in part (d), needing $4(x - 1)$ rather than $4x - 1$ defeated the majority. However, this caused no undermining of confidence as many answered part e(i) well, with sufficient justification to support the given answer.

It should be noted, however, that answers which simply started with $k = \frac{8 \times 5}{4}$ with no evidence of using correct log laws could not earn credit. In part (e)(ii) most candidates

knew they had to solve ' $\frac{5}{4} = 2^{4x-3}$ ' and started by taking logs, but then resorted to calculators rather than establish an exact result (with numbers hinted at in the previous part). The best solution, given by some, was to multiply both sides by ' 2^3 ' and then take logs, to the base 10, of ' $10 = 2^{4x}$ ', which led directly to the result.

Q26.

- (a) This was well answered by the majority of candidates with many fully correct responses. The main error with the stretch was using the $\wedge$ -direction with a scale factor of 2 and with the translation the vector often contained +1.
- (b) This part was well answered by many candidates.
- (c)
 - (i) This was not very well answered by weaker candidates with many just rearranging the expression and then writing down the answer given without any convincing step being shown.
 - (ii) This was answered better than part c(i) with many correct factorisations seen. The main error was with the rejection of the solution $e^{2x} = -1$.
- (d) Some candidates obtained the correct answer and earned full marks. Most candidates made a reasonable attempt at the integrals with reasonable success and most showed enough to earn the B mark for subtraction of their areas. Incorrect limits of 6 and 0 often spoiled what would have been very good attempts.

Q27.

Most candidates were able to score marks in part (a), although the mark in part (a)(iii) was not awarded for answers left as ' $\log_e 5$ '. In part (b), candidates generally scored the method marks but frequently failed to pick up the accuracy mark. The most common wrong answer was '1.66', not always simply from a premature approximation.

Part (c), as expected, proved to be beyond the ability of the average candidate; some did score a mark for ' $p = 2^m$ ', but this was often followed by ' $pq = 2^m \times 8^n = 16^{mn}$ '. Better candidates reached ' $pq = 2^m \times 2^{3n}$ ', but a significant number of these candidates then simplified incorrectly to reach the wrong answer ' $pq = 2^{3mn}$ '.

Q28.

Part (a) was well answered, with most candidates gaining both marks. The most common

error was $3 \ln e^x = \ln 4$, resulting in $x = \frac{\ln 4}{3}$.

Part (b) was generally well done.

Part (c) was not very well answered. Most candidates obtained 1 mark for the y -values but many candidates stopped at this stage and therefore lost the final 2 marks. The error in part (a) was often repeated.

Q29.

The work required to answer parts (a)(i) and (a)(ii) was well understood but a significant number of candidates did not give an explicit value for p or an explicit value for x . Candidates at this level should not be leaving their answers in an embedded form, for example, in part (a)(ii) answers were sometimes left as ' $52 \times 0.75 = 125$ '.

Part (b) was answered better than the corresponding work in previous years and there continued to be an improvement in knowledge and use of the laws of logarithms as seen in solutions to part (c) where many candidates were able to score at least half marks by correctly applying two laws of logarithms. Only more able candidates either used $\log_a a = 1$

or applied $\log_a \frac{x}{k} = -1 \Rightarrow \frac{x}{k} = a^{-1}$ towards expressing x in terms of a correctly.

Q30.

Part (a)(i) was answered correctly by virtually all candidates, and attaching the £ sign to the value of A was condoned.

Most candidates earned 2 marks in part (a)(ii), either for giving an answer for k to a greater accuracy than that stated in the question or, more commonly, for isolating k in an unsimplified form. Many variations on a correct expression were seen, but with most candidates giving the simplest; the 36th root of 7000/A. However, those who did use logarithms usually did it correctly.

In part (b), although most candidates could set up and solve the index equation correctly, the interpretation of the solution showed a common lack of understanding. Although most candidates correctly calculated t to be 56.89, almost all assumed the answer to be 56 and not 57, thus losing the final mark. It seemed that many thought you always round down in this type of question and didn't read the context carefully. Relatively few candidates used a trial and improvement approach, but of those who did many actually demonstrated that the required answer was indeed 57.

Q31.

In part (a), many candidates gave the correct value for where the curve met the y -axis, but a significant minority drew no part of the graph in the second quadrant, their graphs stopping at the point (0, 1). Many candidates scored at least half marks for their descriptions of the geometrical transformations in part (b). The most common wrong answer in part (b)(i) was “a stretch of scale factor 2 parallel to the y -axis”, and in part (b)(ii) the direction of the translation vector caused the most problems.

Part (c)(i) was poorly answered, with many candidates failing to give sufficient justification for the printed answer. A very common error was to write 9^x as $3y$ and yet still reach the correct factorised quadratic equation.

Although part (c)(ii) was answered better, the majority of candidates failed to gain all the four marks. The most common wrong approach was to take logarithms of each term in the given equation. Those candidates applying this incorrect method had clearly not seen the implication of the “Hence”. Those candidates who had seen the link and started with $3^x = 1$ and $3^x = 2$ generally scored all four marks.

Q32.

This question fell into two clear parts: most candidates gained full, or nearly full, marks on part (a), but most were only able to score the first mark in part (b). In part (a)(i), most wrote down the value of A correctly. In part (a)(ii), most candidates obtained an expression for k from which it could be evaluated and so gained the marks. Those who did not give an explicit expression for k lost a mark, and those who just verified that $20k^{60} \approx 2000$ scored no marks. Similarly, candidates were mostly correct in part (a)(iii), but a few made an error in $2008 - 1885$, and some did not round to the nearest 1000.

In part (b), many candidates showed that they had understood the question by setting up the correct equation, but few of those were then able to solve it correctly. The common problem was that, having written down an equation of the form $ak^t = bm^t$, candidates wrote the corresponding log equation as $t \log(ak) = t \log(bm)$ and so, confused by their t apparently cancelling, abandoned their answer. Those who wrote the log equation correctly usually went on to complete the question correctly. The other common mistake was a misunderstanding of the question in which the expression for Q was equated to the result from part (a)(iii); this scored no marks.

Q33.

Candidates generally applied the trapezium rule correctly and gave the final answer to the given degree of accuracy. Many candidates failed to gain full marks in part (b)(ii) as their diagrams did not show four trapeziums. There were also a significant number of candidates who either failed to draw any diagram or who drew the sloping sides of their trapeziums under the curve. Most candidates realised that the geometrical transformation was a stretch, but the wrong answer ‘stretch parallel to the y -axis, scale factor 3’ appeared almost as frequently as the correct answer. A high proportion of candidates showed that they were able to use logarithms to solve the equation $6^{3x} = 84$ but some only gave their answer to two decimal places. The final part of the question was poorly answered, although a minority of candidates did score some credit for partially correct answers, for example, $f(x) = 6^{x+1} - 2$.

Q34.

Many candidates seemed to have little experience of formulating a differential equation as often a derivative was missing from their answer to part (a)(i). Attempts at a derivative often involved variables other than the given x and t . Some candidates interpreted the question as 'find x in terms of t ', and many expressions involving an exponential function were seen. Some candidates, who essentially had a correct differential equation, also thought they were required to solve it. Few candidates actually responded to the word 'decreasing' in the question, and many had a minus sign missing from an otherwise correct expression.

The given values of 500 and 20000 were often misinterpreted in part (a)(ii); the 500 often being taken to mean time. However, many candidates did indicate they had understood 500 to be a rate and 20000 the current value of x , but often also had a spurious t or another term in x in their differential equation. Some had a correct value of k , but negative, not apparently noting $k > 0$ had been given in the question. Relatively few fully correct answers to part (a) were seen.

Most candidates found the value of A correctly in part (b)(i), the common error being an apparent misinterpretation of the given information and finding A to be 700.

Part (b)(i) of the question was generally done well with many fully correct answers, and very few solutions by a trial and improvement approach. The majority of candidates could manipulate the expression and take logarithms correctly. Some algebraic errors were seen; in particular those who divided 1900 by 2000, rather than subtracting it, gained no further credit. Some first subtracted the 2000, and then tried to take logarithms of the expression as it stood, usually with an error. Some candidates incorrectly used base 10 logarithms. In completing the question most candidates rounded down their result to give the correct year. Some, who had made an error, did not round down and so had their year incorrect. Some candidates misinterpreted the last request in the question giving their answer as the 51st or 52nd year, rather than 2059.

Q35.

Part (a)(i) was well answered, with the main error being $\int -e^x dx = -2e^{2x}$. In part (a)(ii) the answer was given. Frequently candidates tried to 'fudge' their solutions to arrive at the correct result, with common error being $4 \ln 2 - 2 - \frac{1}{2} = 4 \ln 2 - \frac{3}{2}$, the given answer.

Part (b)(i) was again well answered by the majority of candidate, although in part (b)(ii) the step from $2x = \ln 4$ to obtain $x = \ln 2$ was often unconvincing. Some candidates answered part (c) very well and gained full credit. The main error was using 8 or -8 as the gradient of the normal, however many candidates earned at least partial marks.

Although correct answers were seen in part (d), they were not common. Many candidates earned the method mark by the substitution of $x = 0$ and some went on to find the area of the triangle. The final mark was often lost through a failure to realise that the area was positive.